

LAB EXPERIMENT RECORD

Set 7: Customer Demographics Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	PRANAYAA S
Register Number	192524110
Date	03.10.2026
Day	Saturday

Aim

To analyze customer demographic data using a bar chart, pie chart, and table in R, and to build an interactive R Shiny dashboard combining all three.

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

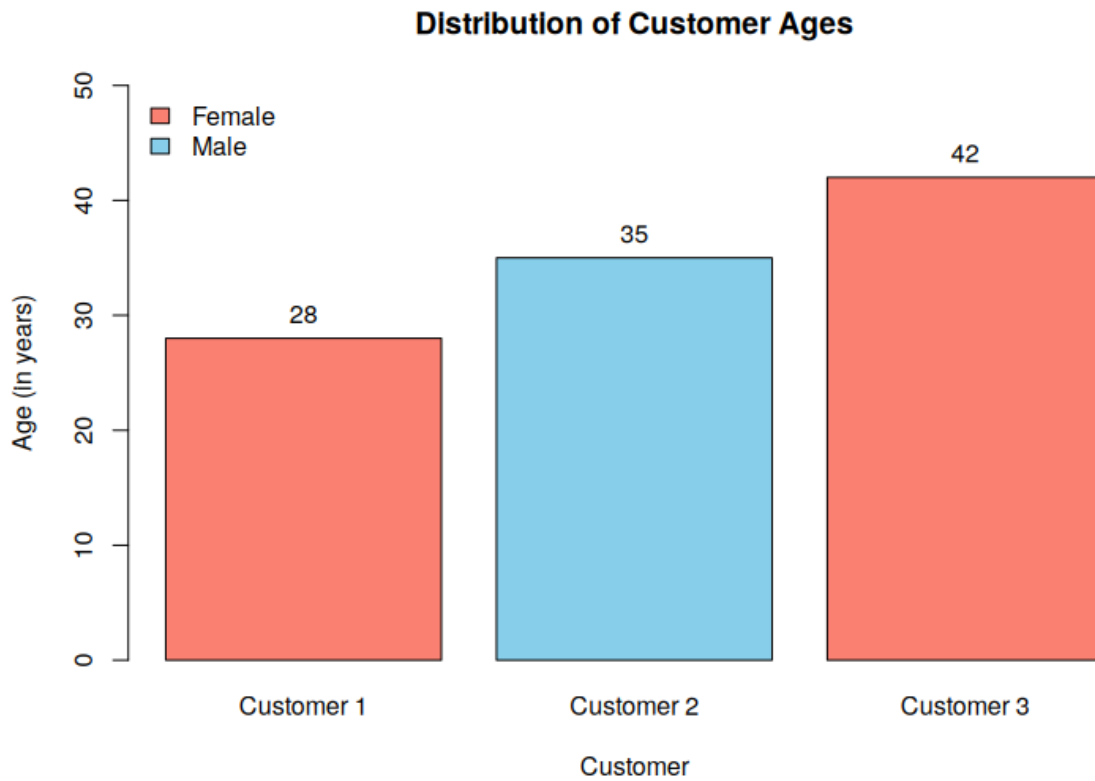
PROGRAM 1: Bar Chart of Customer Ages

Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
# Create dataset for Set 7
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

print(customer_data)

# 1. Bar Chart of Customer Ages
bp <- barplot(customer_data$Age,
  names.arg = paste("Customer", customer_data$Customer_ID),
  col = c("salmon", "skyblue", "salmon"),
  xlab = "Customer",
  ylab = "Age (in years)",
  main = "Distribution of Customer Ages",
  ylim = c(0, 50))
text(bp, customer_data$Age, labels = customer_data$Age, pos = 3)
legend("topleft", legend = c("Female", "Male"),
  fill = c("salmon", "skyblue"), bty = "n")
```



Output 1: Distribution of Customer Ages

Observation: The customers are 28, 35, and 42 years old, with an average age of 35. Customer 3 is the oldest and Customer 1 is the youngest. The ages are evenly spread, with a gap of 7 years between each customer.

PROGRAM 2: Pie Chart of Customers by Gender

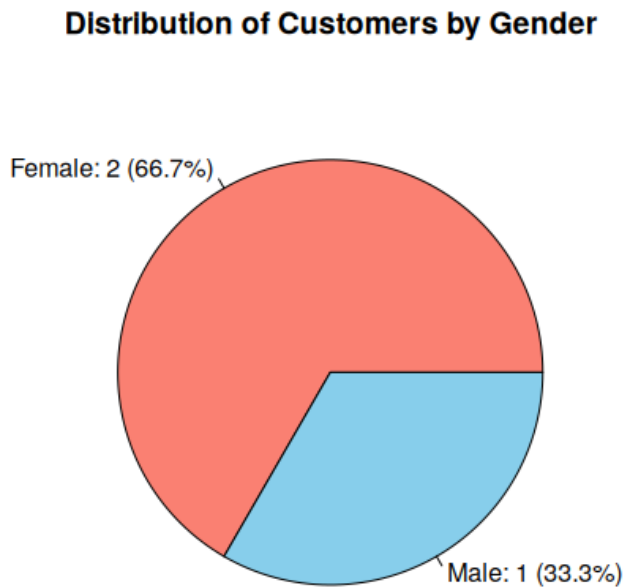
Generate a pie chart to represent the distribution of customers by gender.

```
# Create dataset for Set 7
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

print(customer_data)

# Count customers by gender
gender_counts <- table(customer_data$Gender)
gender_pct <- round(100 * gender_counts / sum(gender_counts), 1)
pie_labels <- paste0(names(gender_counts), ": ", gender_counts, " (", gender_pct, "%)")

# 2. Pie Chart of Customers by Gender
pie(gender_counts,
  labels = pie_labels,
  col = c("salmon", "skyblue"),
  main = "Distribution of Customers by Gender")
```



Output 2: Distribution of Customers by Gender

Observation: Two of the three customers are female (66.7%) and one is male (33.3%), so female customers form the majority of this dataset.

PROGRAM 3: Table of Customer Demographics

Build a table to show the demographic information for each customer. Label the table elements.

```
# Create dataset for Set 7
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

print(customer_data)

# 3. Table of customer demographic information
demo_table <- customer_data
colnames(demo_table) <- c("Customer ID", "Age", "Gender", "Income ($)")

cat("=== CUSTOMER DEMOGRAPHICS TABLE ===\n")
print(demo_table, row.names = FALSE)
```

```
=== CUSTOMER DEMOGRAPHICS TABLE ===
> print(demo_table, row.names = FALSE)
  Customer ID Age Gender Income ($)
1      28 Female    50000
2      35   Male    60000
3      42 Female    75000
>
```

Output 3: Customer Demographics Table

Observation: The table shows the age, gender, and income of each customer with clearly labelled columns. Income increases with age in this data, from \$50,000 for the 28-year-old customer to \$75,000 for the 42-year-old customer.

PROGRAM 4: Interactive Dashboard for Customer Demographics

Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 7
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)
g_cols <- c(Female = "salmon", Male = "skyblue")

ui <- fluidPage(
  titlePanel("Customer Demographics Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("gender", "Gender:", choices = names(g_cols), selected = names(g_cols))
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("agePlot", height = "300px")),
        column(6, plotOutput("piePlot", height = "300px"))
      ),
      h4("Customer Demographics Table"),
      tableOutput("demoTable")
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({ subset(customer_data, Gender %in% input$gender) })

  output$agePlot <- renderPlot({
    d <- filtered()
    barplot(d$Age, names.arg = paste("Customer", d$Customer_ID), col = g_cols[d$Gender],
      ylim = c(0, 50), xlab = "Customer", ylab = "Age (in years)",
      main = "Distribution of Customer Ages")
  }, width = 420, height = 300, res = 96)

  output$piePlot <- renderPlot({
    counts <- table(filtered()$Gender)
    pie(counts, labels = paste0(names(counts), ": ", counts), col = g_cols[names(counts)],
      main = "Customers by Gender")
  }, width = 420, height = 300, res = 96)

  output$demoTable <- renderTable({
    d <- filtered()
    colnames(d) <- c("Customer ID", "Age", "Gender", "Income ($)")
    d
  })
}
```

```

    }, digits = 0)
  }

  shinyApp(ui = ui, server = server)

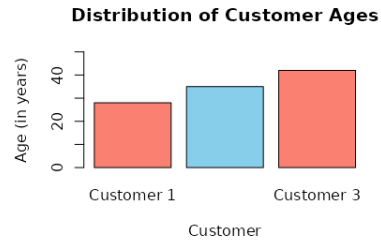
```

Customer Demographics Dashboard

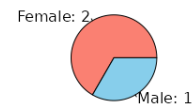
Gender:

☒ Female

☒ Male



Customers by Gender



Customer Demographics Table

Customer ID	Age	Gender	Income (\$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Output 4: Customer Demographics Interactive Dashboard (R Shiny)

Observation: The dashboard shows the age bar chart, the gender pie chart, and the demographics table together. Using the Gender filter shows only female or male customers in all three views at the same time.

Procedure

1. Create the customer data frame with Customer ID, Age, Gender, and Income.
2. Use `barplot()` to show the age of each customer and `text()` to add value labels.
3. Count customers by gender using `table()`, calculate percentages, and use `pie()` to draw the labelled pie chart.
4. Rename the columns with `colnames()` and print the labelled demographics table.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The customer demographic data was successfully visualized using a bar chart, pie chart, and table in R, and an interactive R Shiny dashboard was built combining all three views.

Key R Functions Used

Function	Purpose
<code>data.frame()</code>	Creates the customer demographics dataset.
<code>barplot()</code>	Creates the bar chart of customer ages.
<code>text()</code>	Adds value labels to the bars.
<code>table()</code>	Counts customers by gender.
<code>pie()</code>	Creates the gender pie chart.
<code>colnames()</code> / <code>print()</code>	Labels and displays the demographics table.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>renderTable()</code>	Displays the filtered table on the dashboard.
<code>shinyApp()</code>	Runs the interactive dashboard.